

Public Economics

General information

Each exercise set contains selected Discussion Questions from the textbook and Additional Exercises.

Unless specified otherwise, the discussion in the tutorials only covers the Additional Exercises. You can do the questions from the textbook by yourself; solutions are provided. If you have questions, you can ask your tutorial teacher.

Some exercise sets contain more Additional Exercises than can be discussed during the tutorial. At the start of the tutorial, you can tell the tutorial teacher which exercises you would like to discuss.

Practicum 1

Discussion Questions Rosen:

	8th edition	9th & 10th editions
Chapter 3	4, 9, 10	6, 12, 14
Chapter 4	1	1

Additional Exercises:

- 1.1 Two independent villages, Tinytown and Smallville, are located on opposite sides of a river. Currently, there is no bridge connecting the two villages. The villages decide independently on whether to (help) finance a bridge. Building a bridge would cost 100. If both villages help to finance the bridge, they both pay 50; if only one village finances the construction, it has to pay 100. The benefit of a bridge to Tinytown is 60, and to Smallville 75.

- (a) Is it socially efficient to build the bridge? Suppose the two villages decide independently whether to build the bridge. Will the bridge be constructed (determine the Nash equilibrium)?

Now suppose that the benefit of the bridge to Smallville rises to 125.

- (b) Suppose the two villages decide independently whether to build the bridge. Will the bridge be constructed? Which village is best off?

1.2 Thelma and Louise are neighbours. During the winter, a snowplow can clear the street in front of their houses. Thelma's marginal utility (= marginal benefit) from snowplowing services is $12 - Z$ euro, where Z is the number of times the street is plowed. Louise's marginal utility is $8 - 2Z$ euro. The market price per hour equals 16 euro, which is also equal to the marginal cost of snowplowing.

- (a) Sketch the marginal utility curves and the aggregate willingness to pay. Find the efficient level of snowplowing.
- (b) Suppose Thelma and Louise decide independently how many hours of snowplowing to pay for (of course taking the presence of the other into account). What is the Nash-equilibrium outcome?
- (c) How large is the efficiency loss that occurs under private provision (the outcome under b), relative to the socially efficient outcome (the outcome under a)?

Now suppose that the price of snowplowing drops to 8 euro.

- (d) What would happen if the marginal cost of plowing suddenly drops below 8?

1.3 Adam and Britt derive utility from a public good. Adam's total utility derived from consuming Q units of the public good, measured in euro, is given by

$$U_A = \alpha \ln Q$$

Britt's total utility derived from consuming Q units of the public good, measured in euro, is given by

$$U_B = \beta \ln Q$$

Total production cost of the public good are given by $T = cQ$. The market price per unit of the public good is c . Both Adam and Britt maximise their utility, and they decide independently from each other how many units of the public good to buy. Assume that $\alpha > \beta > 0$ and $c > 0$.

- (a) Derive the socially efficient level of public good provision.
- (b) In the Nash-equilibrium, how many units of the public good does Britt buy? How many units does she consume?

Suppose the government provides the public good, and covers the cost by taxing Adam and Britt such that they both pay half of the total cost of providing the public good. However, the government does not know the preferences of Adam

and Britt. The government makes the following proposal: Adam and Britt (independently) state how many units of the public good they prefer. The government then provides the average number of units (and adjusts the tax accordingly). In other words, if Adams states to prefer Q_A units and Britt states to prefer Q_B units, the government provides $\frac{1}{2}(Q_A + Q_B)$ units and both pay a tax of $\frac{1}{2}c\left(\frac{1}{2}(Q_A + Q_B)\right)$

- (c) Determine the Nash equilibrium of statements Q_A and Q_B .
- (d) Compare the outcome of c) to both the socially efficient level (part a) and the outcome in case of private provision (part b). What does this exercise illustrate?

1.4 Consider a (very) small village, where only two families live: family Adams and family Besley. The size of the available land around the village is 5 acres. This land must be divided into space for agriculture, which creates sandwiches s , and a park of size z acres. The park is a pure public good. Each acre agricultural land yields 50 sandwiches.

- (a) Find the production possibilities frontier. What is the Marginal Rate of Transformation between sandwiches and park size?

The two families derive utility from the number of sandwiches and from the size of the park. The utility of Adams and Besley is given by, respectively:

$$U_A = s_A + 20z$$

$$U_B = \frac{1}{2}s_B + 30\sqrt{z}$$

- (b) Derive the Marginal Rate of Substitution between sandwiches and park size for both families. (Hint: MRS is the slope of an indifference curve, which is equal to the ratio of the marginal utilities of z and s .)
- (c) What is the socially efficient size of the park?
- (d) How large would the park be if the two families decide independently about their contribution to park size (assume that they both own half of the land around the village)?

1.5 Suppose there are two individuals, Anne and Ben, who both derive utility from the consumption of a public good and from the consumption of a private good. Anne's utility function is given by

$$U_A = \ln(x_A) + \ln(G_A + G_B)$$

where x_A is the number of units of private goods Anne consumes, G_A is the number of units of the public good bought by Anne, and G_B is the number of units of the public good bought by Ben. Ben's utility function is given by

$$U_B = \ln(x_B) + \ln(G_A + G_B)$$

where x_B is the number of units of the private good that Ben consumes.

Marginal production cost of both goods are constant and equal to their price on the market. The price of the public good equals p_G . The price of the private goods equals $p_x = 1$. As a result, the marginal rate of transformation between the public good and the private good equals p_G . Anne and Ben have the same level of income y , which they spend fully on the two goods. Given $p_x = 1$, this implies that $x_A = y - p_G G_A$ and $x_B = y - p_G G_B$. Both individuals maximise their own utility.

- (a) Consider the assumptions of this exercise. How can you see in this exercise that the public good is non-rival while the private good is rival?
- (b) Derive that for Anne and Ben, the marginal rate of substitution between the public good and the private good are given by

$$MRS_A = \frac{x_A}{G_A + G_B} \text{ and } MRS_B = \frac{x_B}{G_A + G_B}, \text{ respectively.}$$

- (c) Determine the socially efficient level of units of the public good $G_A + G_B$.

Suppose that the public good is provided by the market. Anne and Ben decide independently how many units of the public good to buy (of course taking the presence of the other into account).

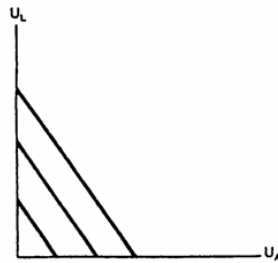
- (d) Determine the Nash-equilibrium that arises on the market (in terms of provision of the public good by Anne and by Ben).
- (e) Explain why the answers to c) and d) are different.

Answers to exercises week 1

The solutions on the Discussion Questions (DQ) in Rosen & Gayer are numbered as follows: Rosen [number DQ in 8th edition] / [number DQ in 9th & 10th editions]. So, Rosen 3.4 / 3.6 gives the solution to DQ 3.4 in the 8th edition and to DQ 3.6 in the 9th & 10th editions of the textbook.

Rosen 3.4 / 3.6:

- a) The social indifference curves are straight lines with slope of -1 . As far as society is concerned, the “util” to Augustus is equivalent to the “util” to Livia.
- b) Social indifference curves are straight lines with slope of -2 . This reflects the fact that society values a “util” to Augustus twice as much as a “util” to Livia.
- c) See the figure.



c.

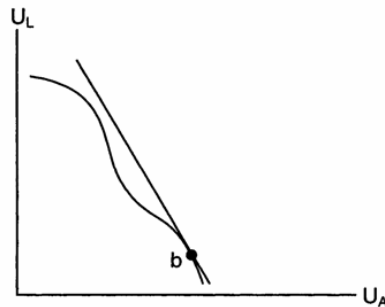
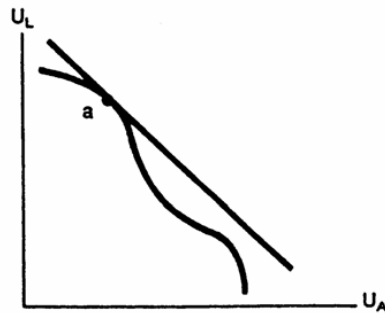


Figure 3.4

Rosen 3.9 / 3.12

Although Victoria's marginal rate of substitution is equal to Albert's, these are not equal to the marginal rate of transformation and the allocation is, therefore, Pareto inefficient. Both people would give up 2 cups of tea for 1 crumpet but, according to the production function, could actually get 6 crumpets by giving up 2 cups of tea. By giving up tea and getting crumpets through the production function, both utilities are raised.

Rosen 3.10 / 3.14

- a) False. As shown on page 41 of the text, equality of the marginal rates of substitution is a necessary, but not sufficient, condition. The MRS for each individual must also equal the Marginal Rate of Transformation.
- b) True. As long as the allocation is an interior solution in the Edgeworth box, the marginal rates of substitution must be equal across individuals. This need not be true, however, at the corners where one consumer has all the goods in the economy.
- c) False. If a policy is a Pareto-improvement, it also increases social welfare, but the other way around is not always true. Social welfare may increase by, for instance, redistributing money from rich to poor. Clearly, this is not a Pareto-improvement.
- d, 8th edition) False. Smokers, especially those without children, are worse off if the tax on cigarettes is raised in order to fund education. Social welfare may improve though, through better education and reduced externalities from smoking.
- d, 10th edition) False. See Figure 3.10: a move from point q to point p3 increases efficiency, but reduces the utility of Eve. Hence, it is not a Pareto-improvement. Note that given an allocation within the UPC, there exists **an** allocation on the UPC that yields a Pareto-improvement.

Rosen 4.1 / 4.1:

- a) Wilderness area is an impure public good – at some point, consumption becomes rival; it is, however, nonexcludable.
- b) These television signals are nonrival in consumption. Producers may be able to choose to make the signal either excludable or non-excludable.
- c) Medical school education is a private good.
- d) Television programs are potentially excludable (Pay-tv etc.) and non-rival (given that you have sufficient tv-sets).
- e, 8th edition) An Internet site is nonrival in consumption (unless there is a lot of traffic), and is in principle excludable.
- e, 9th & 10th edition) An ATM machine is excludable. It is non-rival (everyone can use it), unless it is very busy (in which case you may have to wait).

Exercise 1.1

a) It is efficient to build the bridge, as total benefit is $135 > 100$ (= total cost). Both villages decide not to finance the construction however. Consider this payoff matrix, where the first entry in a cell is the payoff to Tynytown, and the second entry the payoff to Smallville:

		S	
		build	not build
T	build	10, 25	-40, 75
	not build	60, -25	0, 0

For both villages, regardless of what the other village decides, it is always best to choose not build. (Recall from Micro: this is a prisoners' dilemma game, 'not build' is here a dominant strategy and the equilibrium (not build, not build) is called a Nash-equilibrium). Note that both villages would be better off if some third party, for instance the government, forced them to contribute both towards the construction of the bridge.

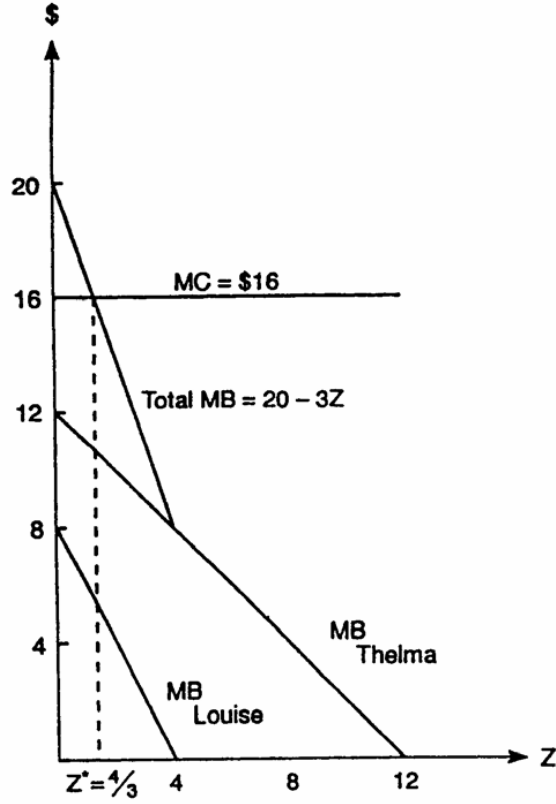
b) Now, Smallville will build the bridge alone. Tynytown is best off. The payoff matrix changes to:

		S	
		build	not build
T	build	10, 75	-40, 125
	not build	60, 25	0, 0

For Tynytown, not build is still the best option, regardless of Smallville's choice (dominant strategy). It follows that Smallville optimally chooses to build.

Exercise 1.2

a) Thelma's marginal benefit is $MB_{THELMA} = 12 - Z$, and Louise's is $MB_{LOUISE} = 8 - 2Z$. The marginal benefit for society as a whole is the sum of the two marginal benefits, or $MB = 20 - 3Z$ (for $Z \leq 4$), and is equal to Thelma's marginal benefit schedule afterwards (for $Z > 4$). The marginal cost is constant at $MC = 16$. Setting $MB = MC$ along the first segment gives $20 - 3Z = 16$, or $Z = \frac{4}{3}$, which is the efficient level of snowplowing.



b) Both Thelma and Louise would not pay for snowplowing: $Z = 0$. Marginal cost is higher than marginal benefit at $Z = 0$ for both.

c) With constant marginal cost, producer surplus is 0. Hence, we have to compute the difference in (total) Consumer Surplus. Under b), CS is 0. Under a), total CS equals the area between the aggregate demand curve and the price. Hence, $CS_a = \frac{1}{2}(20 - 16)\frac{4}{3} = \frac{8}{3}$. Alternative way of calculating is to add the benefits to Thelma and Louise of $Z = \frac{4}{3}$ (area below MB-curves) and subtract the cost:

$$\int_0^{\frac{4}{3}} (MB_T + MB_L - MC) = \int_0^{\frac{4}{3}} (4 - 3Z) = 4\left(\frac{4}{3}\right) - \frac{3}{2}\left(\frac{4}{3}\right)^2 = \frac{8}{3}.$$

One interpretation is: Thelma and Louise's joint valuation of the socially efficient level of provision is $\frac{8}{3}$ euro higher than their joint valuation of the Nash equilibrium outcome.

d) For $MC \leq 8$, it is possible for Louise to credibly free ride, while Thelma provides the efficient allocation. This occurs because if Thelma believes that Louise will free ride, Thelma chooses the amount of Z that maximises her own utility. With $MC \leq 8$, this gives $Z \geq 4$. Yet, for $Z \geq 4$, Louise's marginal benefit is zero, implying that the social MB equals Thelma's MB, see the figure. Hence, if $MC \leq 8$, Louise

free-rides and Thelma pays for $Z \geq 4$, which is socially efficient.

Beyond the question: Note that it depends on the specifics of the exercise that despite free-riding, the outcome is socially efficient. For instance, suppose Louise would have marginal utility of snowplowing equal to $8 - Z$ (rather than $8 - 2Z$), and that marginal cost would be equal to 6. Optimal provision would have $Z = 7$ (As for $Z \leq 8$, social marginal benefit would be $20 - 2Z$). However, if Thelma and Louise decide independently, the only Nash equilibrium is where Thelma pays for $Z = 6$ and Louise contributes nothing. Similar reasoning as above shows that this is indeed a Nash equilibrium: Given that Thelma pays for $Z = 6$, marginal cost are higher than marginal benefit for Louise, so that she optimally contributes nothing. And given that Louise does not contribute, it is optimal for Thelma to pay for $Z = 6$. Hence, the outcome is not socially efficient!

Note that the case where Louise pays for $Z = 2$ and Thelma does not contribute is not a Nash equilibrium: If Thelma would not contribute, Louise optimally pays for $Z = 2$. However, if Louise pays for $Z = 2$, marginal benefits to Thelma are larger than marginal cost ($10 > 6$) so Thelma would contribute something. But then Louise would want to reduce her contributions, etc. Hence, the only Nash equilibrium is where Thelma pays for $Z = 6$ and Louise contributes nothing.

Exercise 1.3

a) $MU_A + MU_B = MC$ gives $\frac{\alpha}{Q} + \frac{\beta}{Q} = c \rightarrow Q = \frac{\alpha+\beta}{c}$. You can also maximise $U_A + U_B - cQ$ with respect to Q .

b). The only Nash equilibrium is where Adam buys $Q = \frac{\alpha}{c}$ and Britt $Q = 0$. Hence, both Adam and Britt consume $Q = \frac{\alpha}{c}$ (note: at $Q = \frac{\alpha}{c}$, we have $MU_B = \frac{\beta}{\alpha/c} = \frac{\beta}{\alpha}c$, which is smaller than c by $\alpha > \beta$. Hence, if Adam buys $\frac{\alpha}{c}$ units, Britt prefers to buy 0 units).

c) The only Nash equilibrium is where $Q_A = 4\frac{\alpha}{c}$ and $Q_B = 0$. As a result, the outcome is $Q = 2\frac{\alpha}{c}$. You can show this by (i) maximising utility of A and B and inferring that $\alpha > \beta$ implies you end up in a corner solution, or by (ii) intuitively arguing that for any Q , the marginal cost are the same for A and B, but the marginal benefits are always higher for A (as $\alpha > \beta$). Hence, if A is content with a particular Q , B prefers a lower level and, hence, sets a lower Q_B . In turn, A sets a higher level.

d) As $\alpha > \beta$, the outcome under c) is higher than the socially efficient outcome. This illustrates that where private provision of public goods tends to yield underprovision (market failure), public provision tends to yield overprovision (government failure). Which of these two is more efficient depends on α and β .

Note 1: Observe that the difference in stated preferences Q_A and Q_B is much larger than the ‘true’ difference in preferred levels of Q .

Note 2: This exercise assumes people have no problem with lying about their preferences. Of course, if all people would be honest about their preferences (even if this

affects their price or tax levels), outcomes would be more efficient.

Exercise 1.4

a) Each acre land can yield 50 s or 1 z . There are 5 acres land, so the maximum amount of sandwiches is 250, and each acre park reduces s by 50. Hence, the PPF equals $s = 50(5 - z)$. It follows that the MRT equals 50. (In general, the MRT is the absolute value of the slope of the PPF, which can be written as $\frac{MC_z}{MC_s}$)

b) MRS between s and z is given by $\frac{\partial U/\partial z}{\partial U/\partial s}$. For family Adams, this gives

$$MRS_A = \frac{20}{1} = 20; \text{ for family Besley, this gives } MRS_B = \frac{\frac{15}{\sqrt{z}}}{\frac{1}{2}} = \frac{30}{\sqrt{z}}.$$

c) Condition for optimal provision of public goods: $MRS_A + MRS_B = MRT$. Substituting gives

$$20 + \frac{30}{\sqrt{z}} = 50$$

Solving for z gives $z = 1$.

A note: in principle, both MRT and MRS could be defined in a reversed manner, so that $MRS = \frac{\partial U/\partial s}{\partial U/\partial z}$ rather than $\frac{\partial U/\partial z}{\partial U/\partial s}$. However, this does not work in case of

deriving the efficient level of public goods provision ($\frac{1}{20} + \frac{\sqrt{z}}{30} \neq \frac{1}{50}$ for $z = 1$). This is because we have to add up the marginal utilities of public good consumption across all individuals, i.e. we want to know how many sandwiches the individuals together want to give up for one additional unit of z . With $MRS = \frac{\partial U/\partial z}{\partial U/\partial s} = \frac{MU_z}{MU_s}$, the MRS is the marginal utility of public good z in terms of utility from consuming private good s , which is what we want to know. With $MRS = \frac{\partial U/\partial s}{\partial U/\partial z}$, we get how much of the public good an individual is willing to give up for an additional private good s . Adding this up over individuals is nonsense, because the additional private good can only be consumed by one individual.

Note that if all goods are private goods, it does not matter whether MRS is written as $\frac{\partial U/\partial x}{\partial U/\partial y}$ or as $\frac{\partial U/\partial y}{\partial U/\partial x}$, as long as the MRT is defined in the same way. It follows directly from the equilibrium condition: if $MRS_X = MRS_Y = MRT$, then $\frac{1}{MRS_X} = \frac{1}{MRS_Y} = \frac{1}{MRT}$.

d) Regardless of the decision by family Besley, family Adams' marginal value of an acre Park equals 20 sandwiches, while the marginal cost are equal to 50. Hence, family Adams does not contribute to the park. Knowing this, family Besley maximises their own family's utility with respect to farm size. Hence, they optimally set $MRS_B =$

MRT , which gives $\frac{30}{\sqrt{z}} = 50$. Solving for z gives $z = 0.36$.

Clearly, this is less than optimal (as found under c)). The reason is that neither family takes the benefits of the park to the other family into account.

a) Non-rivalness implies that units consumed by one person can also be consumed by other people. In this exercise, each person's utility does depends on the amount of the public good provided by the other person, but not on the amount of the private good provided (=consumed) by the other person.

b) Using the definition of the MRS, we get $MRS_A = \frac{MU_G}{MU_x}$. Taking derivatives, we get that $MU_G = \frac{1}{G_A + G_B}$ and $MU_x = \frac{1}{x_A}$. Substituting yields $MRS_A = \frac{\frac{1}{G_A + G_B}}{\frac{1}{x_A}} \rightarrow MRS_A = \frac{x_A}{G_A + G_B}$. Derivation of MRS_B is equivalent.

c) Condition for efficiency: $MRS_A + MRS_B = MRT$ (see lecture 1 / chapter 4). Substituting for x_A and x_B into the expressions for the MRS yields $\frac{y - p_G G_A}{G_A + G_B} + \frac{y - p_G G_B}{G_A + G_B} = p_G \rightarrow \frac{2y - p_G G_A - p_G G_B}{G_A + G_B} = p_G \rightarrow 2y = 2p_G (G_A + G_B) \rightarrow (G_A + G_B) = \frac{y}{p_G}$.

d) For both persons, it is optimal to set $MRS = \frac{p_G}{p_x}$. This gives $\frac{y - p_G G_A}{G_A + G_B} = p_G \rightarrow G_A = \frac{y - p_G G_B}{2p_G}$ and $\frac{y - p_G G_B}{G_A + G_B} = p_G \rightarrow G_B = \frac{y - p_G G_A}{2p_G}$. These can be seen as best-response functions. Solving these two equations with two unknowns (via substitution) yields $G_A = \frac{y - p_G \left(\frac{y - p_G G_A}{2p_G} \right)}{2p_G} = \frac{\frac{1}{2}y + \frac{1}{2}p_G G_A}{2p_G}$ so that $G_A \left(1 - \frac{1}{4} \right) = \frac{y}{4p_G} \rightarrow G_A = \frac{1}{3} \frac{y}{p_G}$ which implies $G_B = \frac{1}{3} \frac{y}{p_G}$ and $G_A + G_B = \frac{2}{3} \frac{y}{p_G}$.

e) Clearly, the Nash-equilibrium yields an inefficiently low level of public goods provision. The reason is that each person benefits from the units of the public good provided by the other person (non-rivalness) but none of them takes these benefits to the other into account in deciding how many units to provide themselves (non-excludability). Therefore, total contributions towards the public good are too low.